DERIVATION OF EOM

$$\text{"L"} = LAG = \underbrace{-Lgm(1-\cos\theta)}_{-Lgm + Lgm\cos\theta} + 0.5m\left[\underbrace{\left(\frac{d\theta}{dt}\right)^2}_{(\dot\theta)^2} L^2 + \underbrace{\Omega^2(w + L\sin\theta)^2}_{\substack{= \Omega^2(w + L\sin\theta)(w + L\sin\theta) \\ = \Omega^2(w^2 + 2wL\sin\theta + L^2\sin^2\theta) \\ = \Omega^2 w^2 + 2\Omega^2 wL\sin\theta + \Omega^2 L^2\sin^2\theta}}\right]$$

$$\text{EOM: } \frac{\delta L}{\delta\theta} = \frac{d}{dt}\left(\frac{\partial L}{\partial\dot\theta}\right) - \frac{\partial L}{\partial\theta}$$

$$\frac{\partial L}{\partial\theta} = -Lgm\sin\theta + 0.5m\left(2\Omega^2 wL\cos\theta + \Omega^2 L^2 \cdot 2\sin\theta\cos\theta\right)$$

$$\Rightarrow \frac{\partial L}{\partial\theta} = -Lgm\sin\theta + m\Omega^2 wL\cos\theta + m\Omega^2 L^2\sin\theta\cos\theta$$

$$\frac{\partial L}{\partial\dot\theta} = 0.5m\left[2\dot\theta \cdot L^2\right] = L^2 m\dot\theta$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot\theta}\right) = L^2 m\ddot\theta$$

$$\Rightarrow \frac{\delta L}{\delta\theta} = L^2 m\ddot\theta - \left(-Lgm\sin\theta + m\Omega^2 wL\cos\theta + m\Omega^2 L^2\sin\theta\cos\theta\right)$$

$$= L^2 m\ddot\theta + Lgm\sin\theta - m\Omega^2 wL\cos\theta - m\Omega^2 L^2\sin\theta\cos\theta$$

$$\Rightarrow \boxed{\frac{\delta L}{\delta\theta} = L^2 m\ddot\theta + Lgm\sin\theta - \Omega^2 Lm\cos\theta(w + L\sin\theta)}$$